MPG No. 6.4.123 2′2′2 (22'2' setting) [Type III, orthorhombic] [T tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} T_x & 0 & 0 \end{bmatrix}$$

$$T_x = \mathbb{T}_x^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{yz} \\ 0 & T_{yz} & 0 \end{bmatrix}$$

$$T_{yz} = \mathbb{T}_{yz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{yz} \\ 0 & -T_{yz} & 0 \end{bmatrix}$$

$$T_{yz} = \mathbb{M}_x^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} T_{xxx} & 0 & 0 \\ T_{yyx} & 0 & 0 \\ T_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & T_{zzx} \\ 0 & T_{yyx} & 0 \end{bmatrix}$$

$$T_{xxx} = 3\mathbb{T}_x^{(1)} + 2\mathbb{T}_x^{\alpha(1)}$$
$$T_{yyx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} + \mathbb{T}_x^{\beta(1)}$$
$$T_{zzx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} - \mathbb{T}_x^{\beta(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ T_{yyx} & 0 & 0 \\ T_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{T_{zzx}}{2} \\ 0 & -\frac{T_{yyx}}{2} & 0 \end{bmatrix}$$

$$T_{yyx} = 2\mathbb{M}_{yz}^{(1)} + 2\mathbb{T}_x^{(2)}$$
$$T_{zzx} = -2\mathbb{M}_{yz}^{(1)} + 2\mathbb{T}_x^{(2)}$$
$$\frac{T_{zzx}}{2} = -\mathbb{M}_{yz}^{(1)} + \mathbb{T}_x^{(2)}$$
$$\frac{T_{yyx}}{2} = \mathbb{M}_{yz}^{(1)} + \mathbb{T}_x^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{zxz} \\ 0 & T_{xyy} & 0 \end{bmatrix}$$

$$T_{zxz} = \mathbb{M}^{(2)}_{yz} + \mathbb{T}^{(3)}_{x}$$
$$T_{xyy} = \mathbb{M}^{(2)}_{yz} - \mathbb{T}^{(3)}_{x}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & T_{yyyz} & 0 & 0 \\ 0 & 0 & 0 & T_{zzyz} & 0 & 0 \\ T_{xxyz} & T_{yyyz} & T_{zzyz} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{zxxy} \\ 0 & 0 & 0 & 0 & T_{zxxy} & 0 \end{bmatrix}$$

$$T_{xxyz} = 2\mathbb{T}^{\beta(1)}_{4x} + \mathbb{T}^{(1)}_{yz}$$
$$T_{yyyz} = \mathbb{T}^{\alpha(1)}_{4x} - \mathbb{T}^{\beta(1)}_{4x} + \mathbb{T}^{(1)}_{yz} + 2\mathbb{T}^{(2)}_{yz}$$
$$T_{zzyz} = -\mathbb{T}^{\alpha(1)}_{4x} - \mathbb{T}^{\beta(1)}_{4x} + \mathbb{T}^{(1)}_{yz} + 2\mathbb{T}^{(2)}_{yz}$$
$$T_{zxxy} = 2\mathbb{T}^{\beta(1)}_{4x} + \mathbb{T}^{(2)}_{yz}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & T_{yyyz} & 0 & 0 \\ 0 & 0 & 0 & T_{zzyz} & 0 & 0 \\ -T_{xxyz} & -T_{yyyz} & -T_{zzyz} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{zxxy} \\ 0 & 0 & 0 & 0 & -T_{zxxy} & 0 \end{bmatrix}$$

$$T_{xxyz} = -4\mathbb{M}^{\beta(1)}_{x} + \mathbb{T}^{(3)}_{yz}$$
$$T_{yyyz} = 2\mathbb{M}^{(1)}_{x} - 2\mathbb{M}^{\alpha(1)}_{x} + 2\mathbb{M}^{\beta(1)}_{x} + \mathbb{T}^{(3)}_{yz}$$
$$T_{zzyz} = -2\mathbb{M}^{(1)}_{x} + 2\mathbb{M}^{\alpha(1)}_{x} + 2\mathbb{M}^{\beta(1)}_{x} + \mathbb{T}^{(3)}_{yz}$$
$$T_{zxxy} = -\mathbb{M}^{(1)}_{x} - 4\mathbb{M}^{\alpha(1)}_{x}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{zxxy} \\ 0 & T_{zxxy} & 0 \end{bmatrix}$$

$$T_{zxxy} = 2\mathbb{T}^{(4)}_{yz}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{zxxy} \\ 0 & -T_{zxxy} & 0 \end{bmatrix}$$

$$T_{zxxy} = -\mathbb{M}^{(2)}_{x}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} T_{xxyz} & 0 & 0 \\ T_{yyyz} & 0 & 0 \\ T_{zzyz} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & T_{zxxy} \\ 0 & T_{xyzx} & 0 \end{bmatrix}$$

$$
\begin{aligned}
T_{xxyz} &= \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{(4)} + 2\mathbb{M}_x^{\alpha(2)} \\
T_{yyyz} &= \mathbb{M}_x^{(3)} - \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{zzyz} &= \mathbb{M}_x^{(3)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{yz}^{(5)} \\
T_{zxxy} &= \mathbb{M}_x^{(4)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{yz}^{(5)} \\
T_{xyzx} &= \mathbb{M}_x^{(4)} - \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{yz}^{(5)}
\end{aligned}
$$

* Rank 4 tensor (as).

$$
\begin{bmatrix}
T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & T_{zxxy} \\
0 & 0 & 0 & 0 & T_{xyzx} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{yzxx} &= \mathbb{M}_x^{(5)} + 2\mathbb{M}_x^{(6)} + 2\mathbb{M}_x^{\alpha(3)} \\
T_{yzyy} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(6)} \\
T_{yzzz} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(6)} \\
T_{zxxy} &= \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} + \mathbb{T}_{yz}^{(6)} \\
T_{xyzx} &= \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} - \mathbb{T}_{yz}^{(6)}
\end{aligned}
$$

* Rank 4 tensor (s).

$$
\begin{bmatrix}
0 & 0 & 0 & T_{xxyz} & 0 & 0 & T_{xxzy} & 0 & 0 \\
0 & 0 & 0 & T_{yyyz} & 0 & 0 & T_{yyzy} & 0 & 0 \\
0 & 0 & 0 & T_{zzyz} & 0 & 0 & T_{zzzy} & 0 & 0 \\
T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & T_{zxxy} & 0 & 0 & T_{zxyx} \\
0 & 0 & 0 & 0 & T_{xyzx} & 0 & 0 & T_{xyxz} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{xxyz} &= \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{(4)} + 2\mathbb{M}_x^{\alpha(2)} - 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{xxzy} &= -\mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{(4)} - 2\mathbb{M}_x^{\alpha(2)} - 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{yyyz} &= 2\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{yyzy} &= 2\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} - \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)} \\
T_{zzyz} &= -2\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} - \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)} \\
T_{zzzy} &= -2\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{yzxx} &= 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} - \mathbb{T}_{yz}^{(3)} \\
T_{yzyy} &= -2\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)} - 2\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)} \\
T_{yzzz} &= 2\mathbb{M}_x^{(1)} - 2\mathbb{M}_x^{\alpha(1)} - 2\mathbb{M}_x^{\beta(1)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)} \\
T_{zxxy} &= -\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(4)} - 4\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)} \\
T_{zxyx} &= -\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(4)} - 4\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(5)} \\
T_{xyzx} &= \mathbb{M}_x^{(1)} + \mathbb{M}_x^{(4)} + 4\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(5)} \\
T_{xyxz} &= \mathbb{M}_x^{(1)} - \mathbb{M}_x^{(4)} + 4\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)}
\end{aligned}
$$

* Rank 4 tensor (a).

$$
\begin{bmatrix}
T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & T_{zxxy} & 0 & 0 & T_{zxyx} \\
0 & 0 & 0 & 0 & T_{xyzx} & 0 & 0 & T_{xyxz} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{yzxx} &= \mathbb{M}_x^{(5)} + 2\mathbb{M}_x^{(6)} + 2\mathbb{M}_x^{\alpha(3)} \\
T_{yzyy} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(6)} \\
T_{yzzz} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(6)} \\
T_{zxxy} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\
T_{zxyx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\
T_{xyzx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)} \\
T_{xyxz} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)}
\end{aligned}
$$

* Rank 4 tensor (t).

$$
\begin{bmatrix}
0 & 0 & 0 \\
0 & 0 & T_{yyyz} \\
0 & T_{zzzy} & 0 \\
0 & T_{yyzy} & 0 \\
0 & 0 & 0 \\
0 & 0 & T_{xxyz} \\
0 & 0 & T_{yzzz} \\
0 & T_{zxxy} & 0 \\
0 & 0 & 0 \\
-\frac{T_{yyyz}}{2} + \frac{T_{yyzy}}{2} + T_{zxxy} & 0 & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{yyyz} &= \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{zzzy} &= -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{yyzy} &= \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)} \\
T_{xxyz} &= 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{yzzz} &= -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)} \\
T_{zxxy} &= 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)}
\end{aligned}
$$